

KRANTHI KUMAR REDDY ANNAPUREDDY – SOFTWARE DEVELOPER

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PROJECTS

Core Technologies: Java, Spring Boot, Spring Security, Git, Docker, Maven, Gradle, MySQL, PostgreSQL

Core Tools: IntelliJ, Postman, VSCode, DBeaver, Swagger

Enterprise Shared Library

March 2026 - Present

Designed and deployed a versioned, modular library distributed via GitHub Packages, effectively eliminating redundant boilerplate domain logic and utilities across the entire corporate microservices ecosystem.

- Designed dual-compatibility support tailored for both Spring MVC (Servlet) and Spring WebFlux (Reactive) paradigms.
- Utilized Java 17 records for modern, immutable `@ConfigurationProperties` and response domains.
- Enforced strict library decoupling and thin dependency graphs using `<optional>true</optional>` and `<scope>provided</scope>` tags, preventing dependency bloat and transitive library conflicts in consuming services.
- Leveraged `@ConditionalOnMissingBean` and validation groups (`CreateValidation`, `UpdateValidation`) to guarantee flexible, plug-and-play configuration.
- Authored comprehensive documentation including Architectural Decision Records (ADRs), compatibility matrixes, and installation guides.

Gateway Server

October 2025 - Present

Developed a high-throughput single entry point for client UI traffic utilizing Spring Cloud Gateway and Reactive WebFlux.

- Configured a highly secure `SecurityWebFilterChain` acting as an OAuth2 Resource Server to reject unauthorized or malformed requests before hitting downstream services.
- Integrated a global CORS configuration using a `CorsWebFilter` bean to restrict and sanitize cross-origin traffic, serving as a secure single entry point for the client UI application.
- Implemented a custom `WebFilter` to enforce strict stateful CSRF protection via `CookieServerCsrfTokenRepository`, ensuring token generation on the very first API handshake.
- Designed a `ReactiveJwtAuthenticationConverterAdapter` to dynamically map JWT claims directly into Spring Security authorities.
- Programmed a dynamic `RouteLocator` bean bundled with custom `GatewayFilter` logic for intelligent, decoupled request routing.

Centralized Authentication Server

January 2026 - Present

Built a multi-tenant identity provider serving as the absolute source of truth for user access control.

- Configured `io.jsonwebtoken` to issue highly customized, state-of-the-art secure JWTs with modified claims sections.
- Exposed a public JWKS endpoint (`/.well-known/jwks.json`) to allow internal Resource Servers to autonomously validate incoming cryptographic tokens.
- Implemented strict Multi-Tenancy isolation leveraging Hibernate 6's native `@TenantId` architecture.
- Created a custom `OncePerRequestFilter` to intercept and resolve tenant metadata dynamically from incoming HTTP request headers.

Data Management System

December 2025 - Present

Designed and built a secure, multi-tenant data management platform comprising a private Spring Boot REST API and a dynamic Angular UI, serving as the central hub for data operations across the ecosystem.

- Implemented a robust multi-tenant architecture using a thread-safe `ThreadLocal` context switcher, ensuring total data security and isolation between different corporate clients.
- Disabled Open Session In View (OSIV) to eliminate lazy-loading risks, drastically reducing memory overhead and database connection bottlenecks.
- Streamlined data compliance and tracking by configuring JPA lifecycle hooks (`@PrePersist`, `@PreUpdate`) to automatically log creation and modification metadata.
- Built a clean UI utilizing Angular Materials and integrated modern Angular Signals to instantly update UI lists on data changes (Add/Delete) without forcing heavy, lagging page-refresh requests.
- Developed a global Angular HTTP Interceptor that transparently attaches JWT/CSRF security tokens to requests and handles silent token-refresh cycles behind the scenes when a user's session expires.

Portfolio

June 2025 - Present

A fully automated, containerized full-stack portfolio application showcasing DevOps and deployment best practices.

- Established a robust CI/CD pipeline using GitHub Actions, reducing manual deployment cycles to zero.
- Configured `logback-spring.xml` with a time-based rolling policy to automate log archiving, ensuring system logs are safely preserved in monthly file rotations for long-term troubleshooting and compliance.
- Deployed the production-ready Angular build natively onto GitHub Pages, embedding strict environment configurations for seamless development-to-production transitions.
- Designed highly reusable front-end notification components (Error, Warning) utilized globally across the application layout.
- Strict CORS configured exclusively for the deployed Spring Boot API service to eliminate unauthorized cross-origin requests.

EDUCATION

Master of Engineering in Internetworking

September 2023 - May 2025

Dalhousie University, Halifax, NS

- Relevant Courses: Programming in Internetworking (Python, Docker, CI/CD pipelines).

Bachelor of Engineering in Electrical and Electronics Engineering

June 2016 - September 2020

Anna University, R.M.D. Engineering College, India

- Relevant Courses: Object Oriented Programming (C, C++, Java languages and Algorithms).

EXPERIENCE

Freelancer

September 2025 - October 2025

Software Developer

- Researched and successfully documented a scalable Micro Frontend architecture for an enterprise Angular application utilizing Webpack Module Federation.
- Bridged the gap between design and production by converting static HTML templates from the UI/UX team into dynamic, high-performance Angular components, ensuring 100% alignment with wireframes and business requirements.

SoWeDid Software Solutions, Hyderabad, India

October 2020 - May 2023

Software Developer

- Refactored legacy codebases to Java 8 streams and modern functional paradigms, significantly improving code readability and reducing execution overhead.
- Designed, developed, and optimized high-performance BIRT reports to deliver critical business intelligence insights.
- Accelerated data retrieval times by optimizing Elasticsearch configurations and queries across large-scale datasets.
- Led the front-end migration from AngularJS to Angular 8, seamlessly configuring `proxy.config.json` for seamless local development and setting up robust `i18n` locale translation systems.
- Spearheaded the integration and documentation of Liquibase into the existing application, establishing a standardized database schema migration process for onboarding teams and external developers.
- Enhanced front-end UX by integrating Angular Datatables for advanced server-side pagination, sorting, and dynamic UI filtering.
- Integrated 3rd-party REST APIs, automated custom Quartz cron jobs, and securely implemented the Paytm payment UI on the front-end.
- Configured custom dynamic mail templates within Spring Boot to streamline automated system notifications.
- Collaborated closely within a fast-paced Agile environment, performing end-to-end debugging across both front-end and back-end systems to resolve complex production blockages.